Mean of a RV

- Let X be a RV with probab. distribun $f(x)$. The mean or the expected value of X is —

$$\boxed{\mu = E(x) = \sum_x x f(x)}$$

if x is discrete, and

$$\boxed{\mu = E(x) = \int_{-\infty}^{\infty} x f(x)\, dx}$$

if x is continuous

- Let X be a RV with probability distribun $f(x)$. The expected value of random variable $g(X)$ is

$$\boxed{\mu_{g(X)} = E[g(X)] = \sum_x g(x) f(x)}$$

for discrete x.

$$\boxed{\mu_{g(X)} = E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x)\, dx}$$

for continuous x.

- Let X & Y be the random variables with joint probability distribu^{n} $f(x, y)$. The mean or expected value of the random variable $g(X, Y)$ is

$$\boxed{\mu_{g(X,Y)} = E[g(X,Y)] = \sum_{x} \sum_{y} g(x,y)\, f(x,y)}$$

if X, Y are discrete

$$\boxed{\mu_{g(X,Y)} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y)\, f(x,y)\, dx\, dy}$$